



Laboratory Report #3

Laboratory Exercise 3: Simple Linear Regression Using JASP

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A. Instructions:

Scenario: You are analyzing a motor-driven system. Your goal is to understand how Temperature, Pressure, and Speed influence Output_Power. Use JASP to perform multiple regression and interpret the results.

1. Analyze whether the 2 variables have a relationship.

r	Level of Correlation
0.00 - 0.25	Very Low
0.26 - 0.49	Low
0.50 – 0.69	Moderate
0.70 – 0.89	High
0.90 – 1.00	Very High

2. Open **JASP** and load the file engine_linear_regression.csv.
3. Go to **Regression** → **Linear Regression**.
4. Assign:
- **Dependent Variable:** Fuel_Consumption
 - **Covariate:** Engine_Load
5. Under **Statistics**, check:
- Model fit (R^2 , Adjusted R^2)
 - Confidence intervals
 - ANOVA table
6. Under **Plots**, enable:
- Residual plots
 - Predicted vs. Observed plot
7. Review the model summary:
- Regression equation: Intercept and slope
 - Significance (p-value)
 - R^2 value
8. Export your APA-style report and graphs as PDF.



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Tables

Linear Regression

Model Summary - Output_Power

Model	R	R ²	Adjusted R ²	RMSE
M ₀	0.000	0.000	0.000	2.132
M ₁	0.880	0.774	0.767	1.029

Note. M₁ includes Speed, Pressure, Temperature

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M ₁	Regression	348.206	3	116.069	109.656	< .001
	Residual	101.614	96	1.058		
	Total	449.821	99			

Note. M₁ includes Speed, Pressure, Temperature

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p
M ₀	(Intercept)	31.941	0.213		149.848	< .001
M ₁	(Intercept)	-5.753	2.304		-2.497	0.014
	Speed	0.013	0.001	0.523	10.784	< .001
	Pressure	3.173	0.318	0.484	9.981	< .001
	Temperature	0.121	0.011	0.543	11.185	< .001

Coefficients Covariance Matrix

Model		Speed	Pressure	Temperature
M ₁	Speed	1.368×10 ⁻⁶	1.235×10 ⁻⁷	2.241×10 ⁻⁷
	Pressure		0.101	1.270×10 ⁻⁴
	Temperature			1.178×10 ⁻⁴

Note. The intercept model is omitted, as no meaningful information can be shown.

Analysis:

- **Write the full regression equation with coefficients.**
The full regression equation from the coefficients table (Model M₁) is Output_Power = -5.753 + (0.013 x Speed) + (3.173 x Pressure) + (0.121 x Temperature). This equation predicts the output power based on the 3 variables given.
- **Which predictor has the strongest influence on output power?**
Based on the standardized coefficients table, temperature has 0.543, speed has 0.523, and pressure has 0.484. The temperature is the highest, which is 0.543, which has the strongest influence on output power.
- **Are any predictors statistically insignificant? What might that suggest?**
Upon checking the p-values, the speed, pressure, and temperature has successfully satisfied the condition, p < 0.001, which makes all of the predictors statistically significant, meaning each variable really does affect the output power.
- **Is the model suitable for predicting system behavior?**
Yes, the model is suitable for predicting system behavior because it is based on the result from R², which is 0.774 or 77.4% of output power variation. Prediction error is much smaller too because of the RMSE drop from 2.132 to 1.029.

Video Link: [JARDELEZA DSL Laboratory 4.mp4](#)



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